Government Engineering College, Siwan

B.Tech 1st Semester Mid-Examination 2025

1st Sem [CSE & CSE(IOT)]

Sub: Basic Electronics Engineering
Sub code: 100114

Time:1hr 30 min
Full Marks: 20

Instructions:

(i) The marks are indicated in right hand margin.
(ii) Attempt ***four*** questions in all.
(iii) Question No. **1** is compulsory.

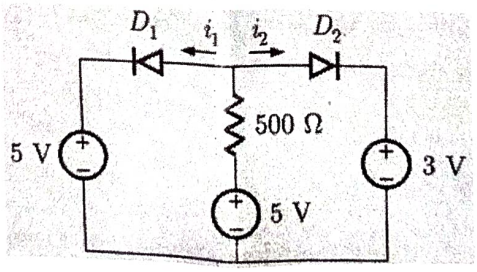
1. Short answer type questions, answer any four questions. [2x4=8]
   (a) What is extrinsic semiconductor ?
   (b) What is thermal voltage at room temperature?
   (c) What is mass action law?
   (d) Draw the VI Characteristics of diode.
   (e) What is biasing of diode ?

2. Explain Energy Band Diagram in details. [4]

3. Explain Half wave rectifier in details. [4]

4. A half-wave rectifier is used to supply 50V d.c. to a resistive load of 800 Ω. The diode has a resistance of 25 Ω. Calculate the a.c. voltage required, I_{RMS}, I_{DC}, ripple factor, rectification efficiency and PIV of rectifier. [4]

5. In the circuit shown below, diodes are ideal. Find the value of current i_1 and i_2. [4]



6. Write short notes on any one of below. [4]
   (i) Clipper and Clamper
   (ii) Drift and Diffusion current
   (iii) Bridge Rectifier